

Details of which parts of my thesis are not my own work

In the following I will explain which parts of my thesis are not solely my own work. These parts constitute a minor component of my thesis.

Chapter 3

Graphene substrates were provided by Graphenea.

Chapter 4

Gold and graphene micro-Hall sensors were fabricated by Dr. Alex Gee.

Chapter 6

Dr Wenhui Niu synthesised the GNR with maleimide groups. Dr Simen Sopp performed the Franck–Condon analysis on temperature data I collected. Mr Fanmiao Kong performed the phonon dispersion calculations. The numerical code through which we simulated the stability diagram and the vibrational states using the rate-equation model was adapted from a numerical code provided by Dr. Jakub Sowa. Dr. Alex Gee helped in collecting more data when I was compelled to stay out of the lab after contracting Covid.

Chapter 7

Dr. Alicia Gotz synthesised the GNRs–OMe. Dr Qiang Chen synthesised the porphyrin–GNR Ni hybrids. Mr Fanmiao Kong assisted me in setting up the electronic calculations on the ARC supercomputing centre.

Appendix A

Dr Keith Andrews synthesised the Dy–Tb spin valve molecule. Dr Alex Gee collected a few of the dataset presented.

Alessandro Lodi

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